Comprehensive Academic and Operational Analysis of the KhmerTutorial Online Platform: A Seven-Chapter Research Framework

The formal seven-chapter research structure provides a standardized academic architecture designed to systematically investigate complex socioeconomic and technological phenomena[1]. By guiding inquiries through background formulation, literature synthesis, theoretical modeling, empirical methodology, data presentation, critical discussion, and strategic execution, this framework translates empirical observations into actionable academic and practical outcomes[1]. This report applies this seven-chapter research methodology to evaluate the operational architecture, financial sustainability, and market validation of KhmerTutorial Online—a specialized single-language micro-tutoring platform connecting Cambodian undergraduate tutors with international learners[4].

Front Matter

The front matter establishes the structural integrity of an academic dissertation, encompassing the title page, table of contents, list of tables, list of figures, acknowledgments, and abstract[1]. Within this structure, the abstract functions as an executive summary of the entire study, providing a dense, multi-perspective overview of the research problem, theoretical underpinnings, empirical methods, key findings, and strategic conclusions[1].

Abstract

The socio-economic environment for Cambodian higher education students is characterized by persistent work-study conflicts, where traditional fixed-shift employment severely disrupts academic continuity[5]. Survey data indicates that 34.5% of undergraduates require flexible remote income, while 37.9% experience acute academic anxiety stemming from rigid work schedules that force them to forfeit class attendance[5]. Simultaneously, foreign expatriates, non-governmental organization (NGO) personnel, researchers, and digital nomads seeking conversational Khmer language skills encounter significant market friction, faced with either costly institutional centers or an absence of dedicated native tutors on broad global EdTech marketplaces[4]. This study evaluates the operational and strategic architecture of KhmerTutorial Online, a two-sided niche EdTech marketplace designed to address these dual market inefficiencies[4]. Utilizing an exploratory sequential mixed-methods design comprising a

quantitative survey of undergraduate supply (80.2% aged 18–20; 44.8% in academic Years 2 and 3) and a four-phase manual Minimum Viable Product (MVP) pilot ($n = 5$ tutor-learner pairs), the study validates platform feasibility, payment clearing mechanisms across borders, and scheduling elasticity[4]. The findings demonstrate that 1-hour micro-tutoring slots effectively eliminate structural work-study conflicts for student tutors while delivering practical, conversational instruction to foreign learners[5]. Furthermore, the platform's architectural design successfully bridges the cross-border payment gap between foreign payment gateways (Stripe/PayPal) and Cambodian settlement networks (ABA PayWay/Wing/Bakong)[4].

Chapter 1: Introduction

1.1 Background

The global educational technology (EdTech) sector has experienced rapid growth, transforming language acquisition into an accessible digital service[4]. However, mainstream global marketplaces prioritize high-density commercial languages, creating significant search friction and quality variance for lower-density regional languages[4]. In Cambodia, higher education undergraduates face substantial economic pressures, requiring supplemental income to fund tuition fees and daily living costs[5]. The local labor market predominantly offers rigid shift work in retail, hospitality, or late-night services, which directly conflicts with daytime university course schedules[5]. The emergence of digital micro-labor platforms presents an opportunity to restructure student employment into flexible, on-demand teaching slots that align with academic commitments[4].

1.2 Problem Statement

Cambodian undergraduate students face a structural economic gap that forces a trade-off between degree completion and financial survival[5]. Quantitative evidence indicates that 34.5% of students require flexible remote earnings, while 37.9% report severe academic anxiety directly linked to rigid employment hours that compel them to miss university lectures[5]. Concurrently, international learners—including NGO staff, embassy diplomats, researchers, and relocating remote workers—confront a fragmented language acquisition market[4]. Existing options are limited to expensive formal language centers that emphasize rigid grammar or global

EdTech platforms that lack curated, native Khmer instruction[4].

1.3 Research Questions

1. How does an on-demand micro-tutoring marketplace model reduce academic anxiety and work-study conflicts among Cambodian undergraduate students[4]?
2. What technical, structural, and user-experience (UX) features are required to optimize cross-border interaction, timezone synchronization, and payment clearing between foreign learners and Cambodian student tutors[4]?
3. To what extent does a hyper-curated single-language niche marketplace outperform generic global EdTech platforms in facilitating localized conversational fluency[4]?

1.4 Objectives

- To measure the socioeconomic and academic impacts of 1-hour micro-tutoring slots on Cambodian university students aged 18–20 during their critical second and third academic years[5].
- To architect and validate a functional web platform incorporating timezone auto-conversion, secure dual-role onboarding, and cross-border financial settlement[4].
- To formulate a sustainable unit economic model that balances learner willingness-to-pay (WTP) with competitive hourly earnings for student tutors[5].

1.5 Significance of the Study

This study contributes to the literature on EdTech platform design, micro-labor dynamics in emerging markets, and language acquisition[4]. By establishing a peer-to-peer micro-tutoring model, the research demonstrates how digital platforms can provide safe, flexible, and dignity-preserving remote employment for vulnerable undergraduate populations in developing nations while fulfilling targeted international demand[4].

Chapter 2: Literature Review

2.1 Theme One: EdTech Marketplaces and Niche Hyper-Curation

Traditional online language learning literature focuses primarily on platform scale, network size, and catalog breadth[4]. However, broad multi-language marketplaces often introduce high search friction and variable instructional quality for low-density regional languages[4]. In contrast, single-language niche marketplaces operate through hyper-curation, establishing rigorous local

tutor vetting, localized curriculum standards, and specialized matching mechanics[4]. This targeted operational focus lowers cognitive friction for foreign learners and creates defensible market positioning against established global competitors[4].

2.2 Theme Two: Undergraduate Work-Study Conflict and Economic Vulnerability

In developing economies, higher education persistence is tightly coupled with concurrent labor force participation[5]. Empirical studies show that rigid, full-time or fixed-shift employment degrades academic performance, elevates drop-out rates, and induces chronic psychological distress among university students[5]. Micro-tutoring platforms introduce temporal elasticity, allowing student workers to monetize unutilized time slots between academic lectures[5]. This micro-earning approach enhances income security while preserving student engagement in core degree programs[5].

2.3 Theme Three: Cross-Border Digital Micro-Labor and Financial Infrastructure

The expansion of international digital service export is constrained by financial rail fragmentation[4]. While international consumers predominantly utilize credit card processing networks and global digital wallets such as Stripe and PayPal, service providers in developing economies depend on domestic clearinghouses and mobile banking systems such as ABA PayWay, Wing, and the National Bank of Cambodia's Bakong network[4]. Technical literature emphasizes that cross-border service platforms must bridge these divergent monetary rails to ensure affordable, real-time settlement for local workers[4].

2.4 The Gap: Application of Miles' (2017) Research Gap Taxonomy

To systematically position this study within the existing literature, the analysis applies Miles' (2017) taxonomy of research gaps, which synthesizes earlier frameworks from Robinson et al. (2011) and Müller-Bloch & Kranz (2014)[7].

Research Gap Type (Miles, 2017)	**Academic Definition**	**Identified Literature Void**	**Application to KhmerTutorial Online Study**
Evidence /	Empirical findings	Global EdTech	Evaluates whether

Contradictory Gap	across existing studies present conflicting conclusions[7].	literature asserts gig platforms empower workers, whereas labor studies highlight wage degradation[7].	specialized micro-tutoring protects student earning potential without inducing labor exploitation[4].
Knowledge Void Gap	Desired empirical data or findings do not exist in current literature[7].	Absence of empirical studies evaluating digital micro-labor platforms tailored specifically for Southeast Asian niche languages[7].	Provides original empirical data on Khmer teaching supply and international learner demand[4].
Practical-Knowledge Conflict Gap	Professional practices contradict published research or lack formal academic analysis[7].	Language centers enforce multi-year formal grammar, whereas expatriates rely on informal, situational conversational practice[5].	Tests practical, module-based conversational training designed for immediate real-world usage[5].
Methodological Gap	Existing research designs distort findings or lack contextual validity[7].	High reliance on macro-level quantitative surveys of gig workers, omitting micro-level	Integrates quantitative demographic surveys with a four-phase manual

		pilot experimentation[7].	MVP pilot[4].
Empirical Gap	Prior theoretical claims require real-world empirical verification[7].	Propositions regarding foreign learner willingness-to-pay for uncertified peer tutors remain unverified in niche markets[7].	Measures willingness-to-pay and satisfaction through live cross-border transactions[4].
Theoretical Gap	Existing theoretical frameworks fail to explain emerging phenomena[7].	Two-sided platform theory rarely models extreme financial rail asymmetries across international borders[7].	Extends platform mediation theory to include asymmetric cross-border banking settlement[4].
Population Gap	Specific demographic sub-groups are underrepresented in prior research[7].	Cambodian undergraduates (aged 18–20, Years 2/3) are under-researched in EdTech micro-labor literature[5].	Focuses directly on Cambodian Year 2 and Year 3 university students as the primary labor supply[5].

Chapter 3: Theoretical and Conceptual Framework

3.1 Theoretical Framework

This study is grounded in three theoretical perspectives:

1. **Two-Sided Platform Theory (Rochet & Tirole)**: Explains how digital intermediaries create value by reducing transaction costs, search friction, and payment asymmetries between distinct user groups[4].
2. **Self-Determination Theory (Deci & Ryan)**: Models human motivation, positing that flexible micro-work enhances autonomy and competence, thereby reducing financial anxiety and supporting academic persistence[5].
3. **Human Capital Theory (Becker)**: Conceptualizes peer tutoring as an asset-building activity where student tutors refine soft skills, English fluency, and cross-cultural communication while continuing their formal degree progression[4].

3.2 Conceptual Framework

The conceptual framework operationalizes platform interactions across three primary dimensions: supply-side student inputs, platform mediating mechanisms, and demand-side learner outcomes[4].

Structural Dimension	Operational Construct	System Function & Empirical Variable
Theoretical Foundation	Formal Academic Theories	Two-Sided Platform Theory, Self-Determination Theory, Human Capital Theory[4].
Supply-Side Inputs	Student Tutor Cohort	Undergraduates aged 18–20 in Years 2 and 3, English fluency, schedule gaps, income need (34.5%)[5].
Mediating Mechanism	KhmerTutorial Platform	Role authentication (JWT/argon2), auto-timezone sync, course

		categorization, payment rails[4].
Demand-Side Outputs	International Learners	Expatriates, NGO staff, digital nomads, conversational fluency, cross-border payment WTP[4].

The interaction across these dimensions demonstrates how platform mediation transforms unutilized student availability into structured micro-tutoring services[4]. By capturing schedule gaps between university classes, the platform generates income for student tutors while delivering flexible, localized instruction to foreign learners[4].

Chapter 4: Methodology

4.1 Research Design

The study adopts an exploratory sequential mixed-methods research design, combining quantitative survey analysis with a four-phase manual Minimum Viable Product (MVP) pilot execution[4].





4.2 Population and Sampling

The research sample was drawn from two target populations:

- **Student Tutor Supply**: Cambodian undergraduates enrolled in universities across Phnom Penh[5]. Sampling criteria prioritized Year 2 and Year 3 students (44.8% of respondents) aged 18–20 (80.2% of respondents) possessing conversational English proficiency and computing hardware[5].
- **International Learner Demand**: Foreign expatriates, NGO workers, embassy personnel, researchers, and digital nomads residing in Cambodia or preparing for relocation[4].

4.3 Technical Architecture and UX Standards

System development adhered to explicit design and architectural specifications prior to prototype implementation[4]:

- **Figma Prototype**: Complete clickable interactive flows covering Home, Search/Browse, Tutor Profile, Booking Calendar, Payment Gateway, Learner Dashboard, and Tutor Dashboard[4].
- **Interface Dynamics**: Motion transitions engineered at $200–300\text{ms}$ ease durations, supported by skeleton loading states and mobile swipeable cards[4].
- **Ergonomic Standards**: Touch targets configured to a minimum 44px tap area, thumb-reachable mobile navigation, and responsive CSS grids[4].
- **Security Architecture**: Role-based access control (RBAC) managed via JSON Web Tokens (JWT) and argon2 password hashing[4].
- **Data Modeling**: Core entities defined for User, TutorProfile, Availability, Booking, and KhmerTeachingFocus (categorized into *Conversational*, *Business Khmer*, *Script/Reading*, and *Travel Khmer*)[4].

4.4 Feature Prioritization and Technical Scope

To manage development risks during a 2-week execution sprint, platform features were classified

into High, Medium, and Low priorities[4]:

Category	Feature Description	Build Priority	Operational Scope & Implementation
Auth & Onboarding	Dual-role signup/login (email/phone)	**High**	Mandatory MVP capability using JWT + argon2 authentication[4].
Auth & Onboarding	Email/phone OTP verification	**Medium**	Secondary security verification layer[4].
Auth & Onboarding	Social OAuth (Google / Facebook)	**Low**	Deferred to future development phases[4].
Tutor Profiles	Bio, photo, hourly rate, focus areas	**High**	Core catalog display for tutor discovery[4].
Tutor Profiles	Video intro upload	**Medium**	Hosted via external media providers[4].
Tutor Profiles	Verification badge / Certifications	**Low**	Trust enhancement scheduled for post-MVP[4].
Search & Discovery	Browse & filter (price, rating, time)	**High**	Primary discovery mechanism for learners[4].

Search & Discovery	Keyword text search	**Medium**	Enables direct bio searching[4].
Search & Discovery	AI-based tutor matching	**Low**	Explicitly excluded from initial development scope[4].
Booking Engine	Weekly availability calendar grid	**High**	Direct slot selection for micro-tutoring[4].
Booking Engine	Timezone conversion algorithm (Δt)	**High**	Automated time offset synchronization[4].
Booking Engine	Low-cost trial lesson flag	**Medium**	Conversion incentive for new learners[4].
Booking Engine	Reschedule and cancellation flow	**Medium**	Automated schedule re-allocation[4].
Payment Gateway	Learner payment (Stripe / PayPal)	**High**	Cross-border inbound currency collection[4].
Payment Gateway	Tutor payout (ABA PayWay / Wing)	**High**	Localized payout processing to Cambodian banks[4].
Payment Gateway	In-app wallet /	**Low**	Excluded from MVP scope[4].

	Lesson credits		
Communication	Booking confirmation email/notifications	**High**	Transactional messaging system[4].
Communication	External video link field (Zoom / Meet)	**High**	Avoids internal WebRTC engineering costs[4].
Communication	In-app real-time messaging	**Low**	Replaced by transactional notifications for MVP[4].
Trust & Reviews	Post-lesson rating & written review	**Medium**	Feedback collection and social proof build[4].
Trust & Reviews	Manual tutor vetting workflow	**Medium**	Administrative quality control layer[4].
Admin Panel	Admin approval and booking audit	**Medium**	Administrative management interface[4].
Design & UX	Complete Figma clickable prototype	**High**	Interactive blueprint for user flow validation[4].

Design & UX	Mobile-first touch-optimized layout	**High**	Responsive layout across mobile viewports[4].

4.5 Data Collection Procedure

Data collection followed a four-step sequential execution roadmap[5]:

1. **Demand Discovery**: Qualitative semi-structured interviews with $15–20$ foreign expatriates and travelers evaluating willingness-to-pay and learning goals[5].
2. **Skills Assessment**: Evaluation of Year 2 and Year 3 undergraduate candidates regarding English communication and instructional readiness[5].
3. **Curriculum Engineering**: Development of 5 core starter modules focusing on high-utility scenarios: *Greetings*, *Transport & Directions*, *Market Shopping & Bargaining*, *Dining Out*, and *Basic Social Interactions*[5].
4. **Manual MVP Pilot**: Pairing 5 tutors with 5 foreign learners conducting live lessons via Zoom, clearing funds manually via ABA PayWay and Bakong digital wallets[4].

4.6 Data Analysis Plan

Quantitative survey data was analyzed using descriptive statistics to map demographic distributions, financial needs, and schedule availability[5]. Qualitative interview transcripts and pilot feedback were coded using thematic content analysis to extract recurring friction points, pedagogical preferences, and user experience requirements[3].

4.7 Validity, Reliability, and Ethical Considerations

Construct validity was established by benchmarking profile fields and booking workflows against established global platforms (Preply) while adapting focus areas to local conditions[4]. Reliability in qualitative coding was maintained through double-blind transcript analysis[5]. Ethical protocols ensured informed consent from all participants, anonymization of survey responses, guaranteed tutor compensation exceeding local service-sector baseline wages, and protections against late-night work hours[4].

Chapter 5: Results and Findings

5.1 Quantitative Results

Quantitative survey analysis provides clear empirical evidence regarding the supply-side student population in Phnom Penh[5].

Quantitative Variable / Metric	Empirical Value	Statistical / Operational Significance
Target Age Cohort (18–20 Years)	80.2%	Confirms high concentration of undergraduate student labor[5].
Academic Progress (Years 2 & 3)	44.8%	Identifies the most academically vulnerable student segment[5].
Need for Flexible Income	34.5%	Demonstrates strong supply-side demand for flexible remote earnings[5].
Reported Academic Anxiety	37.9%	Verifies the psychological impact of rigid shift work on studies[5].
Micro-Slot Duration	1 Hour	Validates the operational feasibility of short teaching slots between classes[5].
Pilot Execution Scale	5 Pairs	Confirms end-to-end operational viability during manual pilot testing[5].

The survey results confirm that Year 2 and Year 3 students experience the highest levels of academic stress[5]. Fixed shift work forces these students to miss mandatory lectures, creating a

compounding cycle of academic vulnerability and financial instability[5].

5.2 Qualitative Findings

Qualitative synthesis of interviews with foreign learners ($n = 15–20$) and post-lesson pilot evaluations revealed three core operational findings[5]:

Finding A: Demand for Practical Conversational Fluency

Foreign learners, particularly NGO personnel and digital nomads, reported frustration with formal language institutes that focus on written script and complex grammar rules during introductory stages[5]. Learners consistently prioritized practical conversational scenarios over formal academic instruction, seeking immediate capability in daily interactions such as directing tuk-tuks, ordering food, and shopping at local markets[5].

Finding B: Cold-Start Liquidity and Trial Lesson Flags

A primary friction point identified during pilot deployment was the marketplace "cold-start problem"—learners hesitated to book unrated tutors with zero platform history[4]. Incorporating a low-cost trial lesson flag significantly reduced initial booking resistance, increasing conversion rates and enabling new tutors to build early social proof through verified post-lesson reviews[4].

Finding C: Infrastructure Integration and Cross-Border Financial Rails

While student tutors possessed adequate computing hardware and smartphone connectivity, cross-border payment transfer emerged as a critical operational vulnerability[4]. Integrating localized payout options (ABA PayWay, Bakong) was necessary to enable tutors to receive earnings directly into local bank accounts without incurring prohibitive international wire fees[4].

5.3 Summary of Sprint Execution Responsibilities

The 2-week execution sprint successfully operationalized these empirical findings across core platform management roles[4]:

- **Chief Executive Officer (CEO)**: Formulated the core strategy, drafted the problem statement using initial research findings, and led pitch narrative development[4].
- **Chief Operating Officer (COO)**: Managed the shared tracking framework (Trello/Notion), audited sprint scope daily, and mitigated execution risks[4].
- **Chief Product Officer (CPO)**: Designed the database schema, configured role authentication (JWT/argon2), built tutor profile creation flows, and implemented timezone

conversion logic[4].

- **Chief Research Officer (CRO)**: Benchmark-priced competitor listings (Preply), recruited pilot student tutors, conducted expat discovery interviews, and formulated cold-start mitigation strategies[4].
- **Chief Financial Officer (CFO)**: Calculated payment gateway fee structures, established cross-border margin spreads, and constructed the platform unit economic break-even model[4].

Chapter 6: Discussion

6.1 Interpretation of Findings

The empirical results demonstrate that KhmerTutorial Online resolves a fundamental structural inefficiency in the Cambodian urban labor market[4]. By decomposing traditional 8-hour shift work into flexible 1-hour micro-tutoring blocks, the platform decouples earning potential from physical presence and rigid temporal schedules[5]. This temporal elasticity directly mitigates the primary driver of student academic anxiety (37.9%), allowing undergraduates to monetize class breaks without compromising degree progression[5].

6.2 Comparison with Existing Literature

The findings extend Two-Sided Platform Theory into hyper-niche regional language markets[4]. While global marketplaces such as Preply rely on massive catalog breadth, they generate high search friction and lack localized onboarding frameworks for tutors in emerging regions[4]. Hyper-curated single-language platforms reduce search costs, improve match quality, and establish higher trust through specialized tutor vetting and tailored starter curricula[4].

6.3 Strategic and Operational Implications

A. Scope Management and Technical Deferrals

To ensure launch readiness within a 2-week development window, explicit scope boundaries were enforced[4]:

Functional Area	MVP Technical Implementation	Explicit Deferral / Future Scope	Strategic Justification for

			Deferral
Video Delivery	External links (Zoom/Google Meet)[4]	Built-in WebRTC video engine[4]	Reduces infrastructure costs and technical complexity[4].
Catalog Scope	Native Khmer language exclusively[4]	Multi-language expansion[4]	Preserves brand focus and concentrates marketplace liquidity[4].
Messaging	Transactional email notifications[4]	Real-time in-app chat engine[4]	Avoids complex socket server maintenance during MVP[4].
Client Application	Mobile-first responsive Web App[4]	Native iOS / Android applications[4]	Avoids dual-codebase overhead and app store review delays[4].
Matching Engine	Filtered search (price, availability)[4]	AI-driven matching algorithms[4]	Unnecessary for initial catalog inventory levels[4].
Dispute Handling	Manual administrative intervention[4]	Automated escrow dispute engine[4]	Manual processing remains cost-effective at low

			initial volumes[4].

B. Platform Feature Framing Matrix

Platform capabilities are categorized by their strategic role in driving marketplace adoption[4]:

Strategic Classification	Target Audience Segment	Feature / Operational Capability	Behavioral Mechanism & Impact
Adoption Barrier Removers	Cambodian Student Tutors	Zero-fee onboarding, flexible hourly scheduling, safe remote work[4].	Eliminates upfront costs, prevents class overlaps, and ensures personal safety[4].
Adoption Barrier Removers	International Learners	Niche curation, auto-timezone sync, low-cost trial lessons[4].	Removes schedule confusion, lowers financial risk, and guarantees native tutors[4].
Value-Add Differentiators	Global Platform Market	Exclusively verified Cambodian student tutors, practical starter modules[4].	Differentiates from generic marketplaces through authentic cultural immersion[4].

C. Unit Economics and Cross-Border Clearing Settlement

The financial architecture relies on managing the margin spread between cross-border collection costs and local disbursement methods[4]:

- **Inbound Gateway Fees**: Processing via Stripe or PayPal incurs international transaction

fees of approximately $2.9\% + \$0.30$ per booking[4].

- **Outbound Payout Settlement**: Disbursements executed via ABA PayWay or Bakong incur minimal domestic transfer costs[4].
- **Platform Take-Rate**: Tiered commission rates (e.g., $15–25\%$) cover transaction fees, hosting infrastructure, and operational overhead while maintaining competitive net earnings for student tutors[4].

Chapter 7: Conclusion and Recommendations

7.1 Summary of the Study

This research evaluated the operational design, theoretical alignment, and execution strategy of KhmerTutorial Online[4]. By addressing the economic needs of Cambodian undergraduates (34.5% income requirement, 37.9% academic anxiety) alongside the practical language requirements of foreign learners, the platform demonstrates a viable model for hyper-niche EdTech marketplaces[4]. The integration of localized starter modules, automated timezone conversion, and cross-border payment rails establishes a scalable framework for digital micro-work in developing markets[4].

7.2 Contributions to Knowledge and Practice

1. **Academic Contribution**: Extends Two-Sided Platform Theory to asymmetrical cross-border labor contexts, demonstrating how specialized intermediaries reduce search friction and transaction costs in low-density language markets[4].
2. **Practical Contribution**: Formulates a validated, low-overhead blueprint for micro-tutoring platforms that offers university students safe, flexible, and income-generating opportunities[4].

7.3 Recommendations for Practice

- **Operational Management**: Maintain strict manual vetting protocols for tutor onboarding to ensure instructional quality before scaling platform automation[4].
- **Curriculum Development**: Expand the initial 5 practical modules into structured, downloadable learning guides to support novice tutors[5].
- **Financial Infrastructure**: Integrate direct NBC Bakong payment APIs to further minimize cross-border processing costs and enable real-time payouts[4].

7.4 Recommendations for Future Research

- **Longitudinal Economic Impact**: Conduct multi-year studies tracking the graduation rates and income trajectories of student tutors participating in digital micro-work[5].
- **Algorithmic Matching**: Evaluate the impact of machine-learning-driven matching engines on booking conversion rates and long-term user retention[4].
- **Geographic Expansion**: Assess the applicability of the hyper-niche single-language marketplace model to other underrepresented language markets in Southeast Asia[4].

7.5 Limitations

The study's findings should be considered within the scope of its operational constraints[5]:

- **Sample Scale**: Pilot validation was conducted with a limited cohort ($n = 5$ pilot pairs; $15–20$ qualitative interviews), requiring larger sample sizes during future scaling[5].
- **Geographic Focus**: Data collection was concentrated among urban university undergraduates in Phnom Penh, which may not fully reflect rural student conditions[5].
- **Infrastructure Assumptions**: The operational model assumes tutors possess functional laptops or smartphones and reliable high-speed internet access[5].

Back Matter

Appendices

Appendix A: Primary Data Collection Instruments

- **Student Survey Questionnaire**: Quantitative survey items measuring demographic characteristics, monthly financial requirements, weekly lecture hours, employment conflicts, and psychometric anxiety indicators[5].
- **Expat Interview Protocol**: Qualitative semi-structured interview guide evaluating prior language learning experiences, monthly training budgets, pain points with formal institutes, and willingness-to-pay for online peer tutoring[5].

Appendix B: Starter Curriculum Specifications

- **Module 1 (Greetings & Social Protocol)**: Introductory greetings, formal titles, and polite conversational framing[5].
- **Module 2 (Transportation & Directions)**: Directing tuk-tuk drivers, navigating urban street layouts, and spatial vocabulary[5].

- **Module 3 (Market Navigation & Bargaining)**: Currency denominations, price inquiries, and negotiation phrasing[5].
- **Module 4 (Dining & Hospitality)**: Ordering local meals, specifying dietary preferences, and payment requests[5].
- **Module 5 (Essential Vocabulary & Assistance)**: Emergency phrases, basic healthcare terminology, and common questions[5].

Appendix C: System Data Schema and Contract Specifications

- **Data Models**: Database schematics defining core system entities:
  - User: id, email, phone, password_hash (argon2), role (TUTOR | LEARNER | ADMIN), created_at[4].
  - TutorProfile: id, user_id, bio, photo_url, hourly_rate_usd, verification_status[4].
  - Availability: id, tutor_id, day_of_week, start_time_utc, end_time_utc[4].
  - Booking: id, learner_id, tutor_id, slot_time_utc, status, meeting_link_url, payment_intent_id[4].
  - KhmerTeachingFocus: id, tutor_id, category (CONVERSATIONAL | BUSINESS | SCRIPT | TRAVEL)[4].

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